CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

Course: MCA412-2 - Research Methodology

LAB Exercise I –

Research Area Identification.

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Domain :

Marine Ecological Informatics & Bio-Optical Remote Sensing

Lab 1

Lab Exercise I:

Research Area Identification.

Marine Ecological Informatics & Bio-Optical Remote Sensing Literature Review

Paper 1: Empirical Relationships for Remote Sensing Reflectance and Noctiluca scintillans Cell Density in the Northeastern Arabian Sea

Sudarsana Rao Pandi, Sanjiba Kumar Baliarsingh, Aneesh Anandrao Lotliker, Nittala S. Sarma, Sarat Chandra Tripathy
Marine Pollution Bulletin, 2020

Objective, Methodology & Key Contributions

This paper focuses on finding a direct link between satellite reflectance (R_{ss}) and the actual cell density of Noctiluca scintillans blooms in the northeastern Arabian Sea. During a cruise on FORV Sagar Sampada, the team collected in-situ optical radiometer profiles alongside surface water samples to measure cell counts, chlorophyll, and suspended matter. Using multiple linear regression, they showed that the band ratio $R_{ss}(531)/R_{ss}(510)$ has a remarkably high correlation with Noctiluca abundance and chlorophyll (adjusted $R^2 = 0.99$). Their key takeaway is that by utilizing the green endosymbiont's peak at 531 nm alongside the 510 nm absorption dip, bloom cell density can be estimated directly without confusing it with other particles.

Observations & Connection to My Research Interest

What I find useful here is that the authors tackle a real challenge in ocean remote sensing: standard blue-green ratios fail when a bloom has photosynthetic endosymbionts living inside it. It reminds me of feature selection in machine learning, where general features can lead to noisy outputs, while a domain-specific band ratio makes the model more reliable. For my project, this paper gives a practical idea for feature engineering during exploratory data analysis (EDA), showing that the 510 nm band is important if I want to isolate Noctiluca blooms in satellite imagery.

Paper 2: Phytoplankton Biomass Dynamics in the Arabian Sea from VIIRS Observations

Wei Shi, Menghua Wang
Journal of Marine Systems, 2022

Objective, Methodology & Key Contributions

The authors set out to map and understand the biannual phytoplankton bloom cycles in the Arabian Sea using multi-year VIIRS satellite data (2012–2019) and historical SeaWiFS records. To explain what drives these blooms, they paired satellite chlorophyll and radiance maps with physical reanalysis data from GODAS, tracking sea surface temperature (SST), mixed-layer depth (MLD), and vertical velocity. Their main contribution was demonstrating two different physical triggers: intense summer coastal upwelling brings cold, nutrient-rich water up from the deep ocean in the western

basin, while winter blooms in the northern basin are caused by surface cooling that deepens the mixed layer to around 80 metres and brings nutrients upward.

Observations & Connection to My Research Interest

This paper helped me understand that analysing satellite colour data alone does not tell the whole story. Chlorophyll is directly connected to physical ocean processes such as vertical advection and convective overturning. Coming from a physics, chemistry, mathematics, and computing background, it makes sense to see how physical ocean parameters explain sudden changes in biological data. It reinforces why my project needs to combine optical data with ocean reanalysis layers such as Copernicus CMEMS and ERA5 to track the environment more effectively.

Paper 3: Recent Changes in Atmospheric Input and Primary Productivity in the North Indian Ocean

J. Kuttippurath, S. Maishal, P. Anjaneyan, N. Sunanda, Kunal Chakraborty
Heliyon, 2023

Objective, Methodology & Key Contributions

This study looked at how primary productivity and chlorophyll changed across the northern Indian Ocean over two decades (1998–2019), and whether airborne dust from surrounding deserts played a major role. The authors combined merged OC-CCI satellite chlorophyll data and CMEMS reanalysis with MERRA-2 aerosol and dust metrics, using HYSPLIT trajectory models to trace where air masses originated. They found an overall decline in net primary productivity across the basin linked to surface warming. They also showed that a 10% to 20% change in air-mass pathways from the Middle East directly affected mineral and nutrient deposition across the Arabian Sea.

Observations & Connection to My Research Interest

I like how this paper looks at the atmosphere as well as the ocean. It shows that desert dust storms can carry trace minerals such as iron and phosphorus, which may trigger unexpected algal blooms even when seasonal upwelling is weak. For my research, this suggests that ocean-colour anomalies are not always caused by processes within the water alone. It gives me a reason to include atmospheric variables as additional features in my data pipeline to help identify sudden bloom events.

Paper 4: Assessing and Improving the Accuracy of Visible Infrared Imaging Radiometer Suite Ocean Color Products in Environments with High Solar Zenith Angles

Hao Li, Xianqiang He, Palanisamy Shanmugam, Yan Bai, Difeng Wang, Teng Li, Fang Gong
Remote Sensing, 2024

Objective, Methodology & Key Contributions

The goal of this paper was to evaluate long-term VIIRS satellite reflectance data (2012–2021) and address the drop in accuracy that occurs when the sun is low on the horizon, at solar zenith angles above 70°. The authors matched global in-situ datasets with satellite passes and developed a neural-network atmospheric correction model (NN-V) to process raw top-of-atmosphere radiances. They showed that standard near-infrared correction algorithms lose almost 50% accuracy at large observation angles, while their neural network reduced relative percentage error from −38.59% to −5.27% and increased valid wintertime data coverage by nearly 20 times.

Observations & Connection to My Research Interest

This paper highlights the importance of data preprocessing. It may be tempting to treat satellite Level-2 products as ground truth, but the study shows that viewing geometry and atmospheric scattering can introduce significant noise, especially in the blue bands. The main computing lesson for me is that quality filtering and geometric checks are essential before running statistical models or other analyses. It also shows how neural networks can help solve complex radiative-transfer problems that traditional models may struggle with.

Paper 5: Global Climate-Change Trends Detected in Indicators of Ocean Ecology

B. B. Cael, Kelsey Bisson, Emmanuel Boss, Stephanie Dutkiewicz, Stephanie Henson
Nature, 2023

Objective, Methodology & Key Contributions

This study examined how to detect climate change in ocean ecosystems, where noisy chlorophyll time series can require more than 30 years of data to reveal a clear trend. Instead of looking at chlorophyll alone, the authors analysed 20 years of MODIS-Aqua multi-band reflectance (R_{ss}) data across the visible spectrum, using multivariate linear regression and the Cochrane–Orcutt method to account for serial autocorrelation. Their key finding was that multi-wavelength reflectance has lower natural noise, revealing significant ecological shifts across more than 40% of the global ocean—mostly in low-latitude areas—with the ocean becoming steadily greener over the two decades.

Observations & Connection to My Research Interest

This paper changed how I think about ocean-colour data. Reducing an entire satellite light spectrum to a single chlorophyll value can discard useful information and increase the effect of noise. Treating reflectance as a multidimensional vector can capture changes in community structure, cell packaging, and detritus together. For my research methodology, a multivariate statistical approach with autoregressive correction seems like a more sensitive way to analyse regional ocean changes than relying only on simple trend lines.